

Hankel Neural Operator

Phase 1: Radial Single-Phase Diffusivity with Heterogeneous Permeability

September 23, 2026

1 Governing equation

Single-phase, slightly compressible radial flow toward a well of radius r_w in a closed circular reservoir of radius r_e , with radially varying permeability $k(r)$:

$$\phi \mu c_t \frac{\partial p}{\partial t} = \frac{1}{r} \frac{\partial}{\partial r} \left(r k(r) \frac{\partial p}{\partial r} \right), \quad r_w \leq r \leq r_e, \quad (1)$$

with $p(r_w, t) = p_{wf}$, $r \partial_r p|_{r_e} = 0$, $p(r, 0) = p_i$.

1.1 Dimensionless form

With a reference permeability k_{ref} , define

$$r_D = \frac{r}{r_w}, \quad t_D = \frac{k_{\text{ref}} t}{\phi \mu c_t r_w^2}, \quad k_D(r_D) = \frac{k(r)}{k_{\text{ref}}}, \quad u = \frac{p - p_{wf}}{p_i - p_{wf}}, \quad r_{eD} = \frac{r_e}{r_w}. \quad (2)$$

Dropping the subscript D on r and t where unambiguous, the problem is

$$\frac{\partial u}{\partial t} = \frac{1}{r} \frac{\partial}{\partial r} \left(r k_D(r) \frac{\partial u}{\partial r} \right), \quad 1 \leq r \leq r_{eD}, \quad (3)$$

$$u(1, t) = 0, \quad (\text{constant bottomhole pressure}) \quad (4)$$

$$r \partial_r u|_{r=r_{eD}} = 0, \quad (\text{no flow}) \quad (5)$$

$$u(r, 0) = 1. \quad (6)$$

Both boundary conditions are homogeneous, which is what allows a basis that satisfies them exactly. The dimensionless well rate is

$$q_D(t) = k_D(1) r \frac{\partial u}{\partial r} \Big|_{r=1}. \quad (7)$$

Phase-1 settings. $r_{eD} = 1000$ (fixed); only $k_D(r)$ varies between samples.

2 Annular finite Hankel transform

2.1 Eigenproblem

Origin: separation of variables. For $k_D \equiv 1$ the spatial operator is $\mathcal{L}u = r^{-1}(ru')'$. Substituting $u(r, t) = \varphi(r)T(t)$ into $u_t = \mathcal{L}u$ gives

$$\frac{T'}{T} = \frac{(r\varphi')'}{r\varphi} = -\lambda^2 \implies T(t) = e^{-\lambda^2 t}, \quad (8)$$

and φ must solve the Sturm–Liouville problem

$$(r\varphi')' + \lambda^2 r \varphi = 0, \quad \varphi(1) = 0, \quad \varphi'(r_{eD}) = 0, \quad (9)$$

with weight r on $[1, r_{eD}]$. Each eigenfunction φ_n is a spatial pressure shape that decays uniformly at rate λ_n^2 : low modes are large-scale and slow, high modes are fine-scale and fast.

Why Bessel functions. Expanding gives $r\varphi'' + \varphi' + \lambda^2 r \varphi = 0$. With $x = \lambda r$ this becomes

$$x^2 \varphi_{xx} + x \varphi_x + x^2 \varphi = 0, \quad (10)$$

Bessel’s equation of order zero, with independent solutions $J_0(x)$ (regular at $x = 0$) and $Y_0(x)$ (singular, $Y_0(x) \sim \frac{2}{\pi} \ln x$ as $x \rightarrow 0$). On a full disk $[0, R]$ the requirement that pressure is finite at $r = 0$ excludes Y_0 , which leaves the classical Fourier–Bessel basis $J_0(j_{0,n}r/R)$. Here the domain begins at the wellbore, $r = 1$, so $r = 0$ is never reached. Y_0 is admissible there, and it is also *required*: J_0 alone cannot satisfy both boundary conditions.

Construction. The general solution is $\varphi(r) = a J_0(\lambda r) + b Y_0(\lambda r)$. The inner condition $\varphi(1) = 0$ requires $a J_0(\lambda) + b Y_0(\lambda) = 0$, which is satisfied by $a = Y_0(\lambda)$, $b = -J_0(\lambda)$:

$$\varphi_n(r) = J_0(\lambda_n r) Y_0(\lambda_n) - Y_0(\lambda_n r) J_0(\lambda_n). \quad (11)$$

The two terms cancel at $r = 1$, so the inner boundary condition holds for *every* λ . Writing $Z_\nu(x) := J_\nu(x)Y_0(\lambda_n) - Y_\nu(x)J_0(\lambda_n)$, we have $\varphi_n(r) = Z_0(\lambda_n r)$ and, using $J'_0 = -J_1$, $Y'_0 = -Y_1$,

$$\varphi'_n(r) = -\lambda_n Z_1(\lambda_n r) = -\lambda_n [J_1(\lambda_n r)Y_0(\lambda_n) - Y_1(\lambda_n r)J_0(\lambda_n)]. \quad (12)$$

The no-flow condition $\varphi'_n(r_{eD}) = 0$ then selects the admissible λ : the inner condition fixes the *shape*, and the outer condition fixes the *spectrum*:

$$\boxed{J_1(\lambda_n r_{eD}) Y_0(\lambda_n) - Y_1(\lambda_n r_{eD}) J_0(\lambda_n) = 0} \quad (13)$$

$\lambda = 0$ is not an eigenvalue: its solutions $a + b \ln r$ are forced to $a = b = 0$ by the two boundary conditions. The roots $0 < \lambda_1 < \lambda_2 < \dots$ are found numerically (bracketing and Brent’s method). For large n , $\lambda_{n+1} - \lambda_n \approx \pi/(r_{eD} - 1)$. The smallest eigenvalue satisfies

$$\lambda_1^2 \approx \frac{2}{r_{eD}^2 (\ln r_{eD} - \frac{3}{4})}. \quad (14)$$

For $r_{eD} = 1000$: $\lambda_1 = 5.688 \times 10^{-4}$, $\lambda_1^2 = 3.235 \times 10^{-7}$ (approximation: 3.248×10^{-7}), so the slowest time scale is $\lambda_1^{-2} \approx 3 \times 10^6$.

2.2 Shape of the eigenfunctions

Near the well ($\lambda_n r \ll 1$). Using $J_0(x) \approx 1$ and $Y_0(x) \approx \frac{2}{\pi} (\ln \frac{x}{2} + \gamma)$,

$$\varphi_n(r) \approx Y_0(\lambda_n) - Y_0(\lambda_n r) \approx -\frac{2}{\pi} \ln r. \quad (15)$$

Every mode begins as the logarithmic steady radial-flow profile $p \propto \ln r$, so the basis contains the near-well physics by construction. For $r_{eD} = 1000$, all of $\varphi_1, \dots, \varphi_4$ agree with $-\frac{2}{\pi} \ln r$ to within 0.2% at $r = 10$. Since $\lambda_1 r_{eD} \approx 0.57$, the first mode stays close to this log profile across most of the reservoir (Fig. 1a).

Far from the well ($\lambda_n r \gg 1$). With $J_0(x) \approx \sqrt{2/(\pi x)} \cos(x - \frac{\pi}{4})$, $Y_0(x) \approx \sqrt{2/(\pi x)} \sin(x - \frac{\pi}{4})$, and writing $Y_0(\lambda_n) = A_n \sin \alpha_n$, $J_0(\lambda_n) = A_n \cos \alpha_n$,

$$\varphi_n(r) \approx A_n \sqrt{\frac{2}{\pi \lambda_n r}} \sin\left(\alpha_n + \frac{\pi}{4} - \lambda_n r\right), \quad (16)$$

a sinusoid in r with amplitude decaying like $r^{-1/2}$. The weight r in the inner product compensates for this decay exactly: $\sqrt{r} \varphi_n / \sqrt{N_n}$ behaves like a normalized sine mode on an interval, with amplitude close to $\sqrt{2/r_{eD}} \approx 0.045$ (Fig. 1b). The zero-slope condition at r_{eD} then explains the eigenvalue spacing $\approx \pi/(r_{eD} - 1)$.

Zeros, sign, and scale. By Sturm–Liouville theory, φ_n has exactly $n - 1$ zeros in $(1, r_{eD})$ (confirmed numerically for $n \leq 4$). Each φ_n is defined only up to a constant factor. With the form above, $\varphi_n < 0$ near the well, which is why the coefficients of $u \equiv 1$ (Section 3.1) are negative. Dividing by N_n in the transform, or using the normalized functions $\varphi_n / \sqrt{N_n}$, keeps everything consistent.

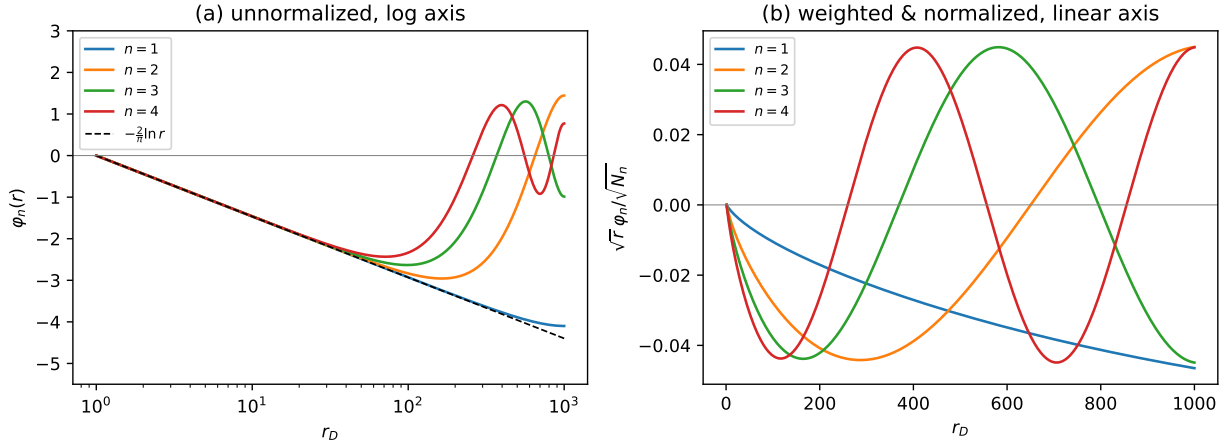


Figure 1: First four annular eigenfunctions for $r_{eD} = 1000$. (a) Unnormalized φ_n on a log axis: all modes follow $-\frac{2}{\pi} \ln r$ near the well and separate further out. (b) $\sqrt{r} \varphi_n / \sqrt{N_n}$ on a linear axis: sine-like modes of nearly equal amplitude with $n - 1$ interior zeros and zero slope at $r = r_{eD}$.

2.3 Orthogonality and normalization

$$\int_1^{r_{eD}} r \varphi_m \varphi_n dr = N_n \delta_{mn}, \quad N_n = \frac{r_{eD}^2}{2} \varphi_n(r_{eD})^2 - \frac{2}{\pi^2 \lambda_n^2}, \quad (17)$$

where the second term uses the Wronskian identity $J_1(x)Y_0(x) - J_0(x)Y_1(x) = 2/(\pi x)$, which gives $Z_1(\lambda_n) = 2/(\pi \lambda_n)$.

2.4 Transform pair

$$\hat{u}_n = \mathcal{H}[u]_n = \frac{1}{N_n} \int_1^{r_{eD}} r u(r) \varphi_n(r) dr, \quad u(r) = \mathcal{H}^{-1}[\hat{u}](r) = \sum_{n=1}^{\infty} \hat{u}_n \varphi_n(r). \quad (18)$$

Every truncated reconstruction $\sum_{n \leq K} \hat{u}_n \varphi_n$ satisfies both boundary conditions exactly.

3 Reference solutions

3.1 Homogeneous case ($k_D \equiv 1$)

Since $\mathcal{L}\varphi_n = -\lambda_n^2 \varphi_n$, the dynamics are diagonal:

$$\hat{u}_n(t + \Delta t) = e^{-\lambda_n^2 \Delta t} \hat{u}_n(t). \quad (19)$$

With $u(r, 0) = 1$ and $\int_1^{r_{eD}} r \varphi_n dr = -2/(\pi \lambda_n^2)$:

$$u(r, t) = - \sum_{n=1}^{\infty} \frac{2}{\pi \lambda_n^2 N_n} \varphi_n(r) e^{-\lambda_n^2 t}, \quad (20)$$

$$q_D(t) = \sum_{n=1}^{\infty} \frac{4}{\pi^2 \lambda_n^2 N_n} e^{-\lambda_n^2 t}. \quad (21)$$

These are the benchmarks for validating the numerical solver. The pressure itself follows from $p(r, t) = p_{wf} + (p_i - p_{wf}) u(r, t)$.

Convergence of the series. Mode n is damped by $e^{-\lambda_n^2 t}$, so the number of modes needed grows as $t \rightarrow 0$. Table 1 lists the number of terms required for an absolute accuracy of 10^{-3} in $u(2, t)$, measured against a 1908-term reference ($\lambda_n \leq 6$).

t_D	$u(2, t_D)$	modes for 10^{-3} accuracy
10^1	0.3687	190
10^2	0.2395	54
10^4	0.1358	5
10^6	0.0808	1

Table 1: Modes needed for the homogeneous series at $r_D = 2$, $r_{eD} = 1000$.

Late times need only a few modes. At early times the pressure front near the well is sharp and needs hundreds. This is the series view of the early-time (Gibbs) difficulty, and it supports pairing $\Delta t = 100$ with $K = 32$ once $t \gtrsim 100$.

Flow regimes. Figure 2 shows the analytic solution. The pressure disturbance spreads outward roughly linearly in $\ln r$ (infinite-acting radial flow). In this period the rate follows the classical approximation $q_D \approx 2/(\ln t_D + 0.809)$ (e.g. ≈ 0.20 at $t_D = 10^4$). Once the disturbance reaches r_{eD} (around $t_D \sim 10^5$ – 10^6), the reservoir depletes and q_D declines exponentially. In that regime the first mode alone, $q_D \approx \frac{4}{\pi^2 \lambda_1^2 N_1} e^{-\lambda_1^2 t}$, reproduces the full series.

3.2 Heterogeneous case

Substitute $u = \sum_n \hat{u}_n \varphi_n$ into (3), project onto φ_m with weight r , and integrate by parts. The boundary terms vanish because $\varphi_m(1) = 0$ and $u'(r_{eD}) = 0$. The result is

$$\frac{d\hat{\mathbf{u}}}{dt} = \mathbf{A}(k_D) \hat{\mathbf{u}}, \quad A_{mn}(k_D) = -\frac{1}{N_m} \int_1^{r_{eD}} r k_D(r) \varphi'_m(r) \varphi'_n(r) dr, \quad (22)$$

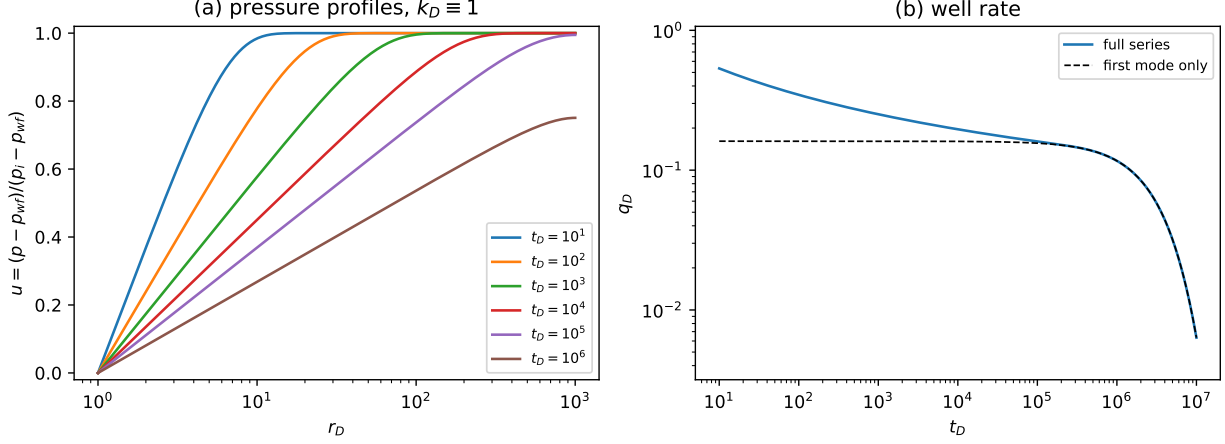


Figure 2: Analytic solution for $k_D \equiv 1$, $r_{eD} = 1000$ (1908 modes). (a) Dimensionless pressure $u = (p - p_{wf})/(p_i - p_{wf})$ at $t_D = 10^1, \dots, 10^6$. (b) Well rate $q_D(t)$: the full series compared with the first mode alone, which matches the boundary-dominated exponential decline.

so the exact one-step map is

$$\hat{\mathbf{u}}(t + \Delta t) = \exp(\Delta t \mathbf{A}(k_D)) \hat{\mathbf{u}}(t). \quad (23)$$

For $k_D \equiv 1$, $\mathbf{A} = -\text{diag}(\lambda_n^2)$ and (23) reduces to (19). For heterogeneous k_D , $\mathbf{A}$ is dense, so permeability couples the modes. The operator to learn is linear in u and nonlinear in k_D : the network must learn $k_D \mapsto \exp(\Delta t \mathbf{A}(k_D))$. This motivates spectral layers that act on the Hankel coefficients.

4 Learning problem

4.1 Operator

$$\mathcal{G} : (u(\cdot, t), \ln k_D(\cdot)) \mapsto u(\cdot, t + \Delta t), \quad (24)$$

with fixed $\Delta t = 100$. Full trajectories are obtained by autoregressive rollout $u^{(j+1)} = \mathcal{G}_\theta(u^{(j)}, \ln k_D)$. Trajectories start at $t_0 > 0$, taken from the reference solver, to avoid the discontinuity between $u(r, 0) = 1$ and $u(1, t) = 0$ (which would cause Gibbs oscillations near the well).

Mode truncation. Mode n is damped per step by $e^{-\lambda_n^2 \Delta t}$. With $\Delta t = 100$ and $\lambda_n \approx n\pi/999$, modes $n \gtrsim 30$ are strongly damped in one step, which motivates $K = 32$.

4.2 Hankel Neural Operator architecture

Inputs on the grid: $a(r) = (u(r, t), \ln k_D(r))$, optionally with the coordinate $\ln r$.

$$v^0(r) = P a(r), \quad P : \mathbb{R}^{d_a} \rightarrow \mathbb{R}^C \text{ (lifting)}, \quad (25)$$

$$v^{\ell+1}(r) = \sigma\left(W_\ell v^\ell(r) + b_\ell + \sum_{n=1}^K R_{\ell,n} \hat{v}_n^\ell \varphi_n(r)\right), \quad \ell = 0, \dots, L-1, \quad (26)$$

$$u(r, t + \Delta t) \approx g(r) Q v^L(r), \quad Q : \mathbb{R}^C \rightarrow \mathbb{R} \text{ (projection)}, \quad (27)$$

where $\hat{v}_n^\ell = \mathcal{H}[v^\ell]_n$ is applied channel-wise, $R_{\ell,n} \in \mathbb{R}^{C \times C}$ are learned *real* spectral weights (the transform is real, unlike the complex weights of FNO), W_ℓ is a pointwise linear map, and σ is GELU. The factor $g(r)$ with $g(1) = 0$ (e.g. $g = \ln r / \ln r_{eD}$) enforces the inner boundary condition exactly. The exact form of this hard constraint is still an open design choice.

4.3 Discretization

Use a log-spaced grid $x_j = \ln r_j = j \Delta x$, $j = 0, \dots, N-1$, $\Delta x = \ln r_{eD} / (N-1)$. Since $r dr = r^2 dx$, the quadrature weights are $w_j = c_j r_j^2 \Delta x$ (trapezoidal c_j). Let $\mathbf{W} = \text{diag}(w_j)$ and define the inverse transform by sampling the eigenfunctions:

$$\mathbf{u} = \mathbf{B} \hat{\mathbf{u}}, \quad B_{jn} = \varphi_n(r_j). \quad (28)$$

Two choices for the forward transform $\hat{\mathbf{u}} = \mathbf{T}\mathbf{u}$ were compared:

$$\text{quadrature: } T_{nj} = \frac{w_j \varphi_n(r_j)}{N_n}, \quad (29)$$

$$\text{least squares: } \mathbf{T} = (\mathbf{B}^\top \mathbf{W} \mathbf{B})^{-1} \mathbf{B}^\top \mathbf{W}. \quad (30)$$

The quadrature form discretizes the continuous transform directly, so $\mathbf{T}\mathbf{B} = \mathbf{I}_K$ holds only up to the $O(\Delta x^2)$ trapezoidal error. The least-squares form is the $\mathbf{W}$ -weighted best fit of $\mathbf{u}$ by K modes. It satisfies $\mathbf{T}\mathbf{B} = \mathbf{I}_K$ exactly, $\mathbf{B}\mathbf{T}$ is a projector, and the exact homogeneous step (19) is reproduced to machine precision. Table 2 shows that it is also far more accurate for functions outside the span of the K modes. **The least-squares form (30) on a log grid ($\alpha = 1$) with $N = 1024$ is adopted.**

$\mathbf{T}$	N	$\max \mathbf{T}\mathbf{B} - \mathbf{I}_K $	rel. error of $\mathcal{H}[1]$
quadrature	512	1.9×10^{-3}	1.8×10^{-3}
quadrature	1024	4.4×10^{-4}	4.2×10^{-4}
quadrature	2048	1.1×10^{-4}	1.0×10^{-4}
least squares	512	9×10^{-16}	9.8×10^{-7}
least squares	1024	2×10^{-15}	1.9×10^{-7}
least squares	2048	2×10^{-15}	4.5×10^{-8}

Table 2: Transform accuracy on the log grid, $K = 32$, $r_{eD} = 1000$ (`phase_1/grid_study.py`). The error of $\mathcal{H}[1]$ is measured against the exact coefficients $-2/(\pi \lambda_n^2 N_n)$, relative to the largest one. A higher-order (Gregory) quadrature was also tried and was less accurate at these N , because products $\varphi_m \varphi_n$ are under-resolved near r_{eD} .

$\mathbf{T} \in \mathbb{R}^{K \times N}$ and $\mathbf{B} \in \mathbb{R}^{N \times K}$ are precomputed once and stored as fixed buffers. A spectral convolution costs $O(NKC + KC^2)$. The reconstruction error of a smooth function satisfying both boundary conditions levels off at $\approx 7 \times 10^{-4}$ for all grids; this is the $K = 32$ truncation error, not a discretization error.

Remark 1 (Resolution near the outer boundary). *The log grid concentrates points near the well, but mode K oscillates with wavelength $2\pi/\lambda_K$ everywhere, and grid spacing grows like $r \Delta x$. Requiring P points per wavelength at $r = r_{eD}$ gives*

$$N \gtrsim \frac{P \lambda_K r_{eD} \ln r_{eD}}{2\pi}. \quad (31)$$

For $K = 32$ ($\lambda_K \approx 0.1$), $r_{eD} = 1000$, $P = 8$: $N \gtrsim 900$. With the least-squares transform this estimate is conservative: $N = 512$ already gives coefficient errors of $\sim 10^{-6}$ (Table 2). It may still apply to the finite-volume solver, which must resolve the solution rather than the modes.

5 Data

Generated by `phase.1/dataset.py`; stored in `phase.1/data/` as `train.npz`, `val.npz` and `test.npz`.

- **Permeability:** $\ln k_D(r) = m + \xi(\ln r) + s\mathbb{I}[r < r_s]$, clipped to $[-3, 3]$. The mean shift is $m \sim \mathcal{N}(0, 0.3^2)$. The field ξ is a zero-mean Gaussian random field in $x = \ln r$ with squared-exponential covariance $\sigma^2 e^{-(x-x')^2/(2\ell^2)}$, $\sigma \sim U(0.5, 1)$, $\ell \sim U(0.3, 1.5)$. With probability 0.5 a near-well skin offset $s = \ln(k_s/k) \sim U(\ln 0.1, \ln 5)$ applies for $r < r_s$, r_s log-uniform in $[2, 20]$. About 3.5% of the fields reach the clip.
- **Reference solver:** vertex-centered finite volumes on the $N = 1024$ log grid, face fluxes exact for u linear in $\ln r$, harmonic-mean face permeability, variable-step BDF2 (largest step 20). For $k_D \equiv 1$ the error against the series of Section 3.1 is $\sim 4 \times 10^{-7}$ for $t_D \geq 10^4$.
- **Trajectories:** from $u = 1$ at $t_D = 0$ (so the surrogate never needs the solver), stored every $\Delta t_D = 1000$ up to 5×10^5 (501 states). With $\Delta t_D = 1000$ and $K = 32$ the exact first step from $u = 1$ is representable to 3×10^{-7} (1.5×10^{-2} at $\Delta t_D = 100$). Training pairs $((u^{(j)}, \ln k_D), u^{(j+1)})$ are formed from consecutive states.
- **Size:** 800 / 100 / 100 trajectories (train / validation / test, seeds 0 / 1 / 2), float32, about 2 GB.

Near-well structure. The flux $rk_D \partial_r u$ varies smoothly in r , so $\partial_r u \propto 1/k_D$ and every near-well permeability feature leaves a kink in u that diffusion does not remove. The $K = 32$ eigenfunctions of the $k_D \equiv 1$ problem have a shortest wavelength of about $2\pi/\lambda_{32} \approx 64$ in r and cannot represent it (Figure 3d–f). At $t_D = 5 \times 10^4$ the relative L_2 error of the 32-mode projection has median 4.5% (3.9% without skin, 5.4% with skin) and 90th percentile 11%; 94% of the residual lies at $r < 10$. For $k_D \equiv 1$ the same error is below 10^{-6} from $t_D = 10^3$. The near-well detail must therefore be carried by the pointwise path of the network, and an FNO-1D on the log grid (modes spread evenly in $\ln r$) may be a strong baseline.

6 Evaluation

- One-step relative error in the weighted norm $\|e\|_r^2 = \int_1^{r_{eD}} r e^2 dr$, i.e. $\|u_\theta - u\|_r / \|u\|_r$.
- Rollout error over a limited horizon (about 100–500 steps).
- Well-rate error in $q_D(t)$.
- Baseline: FNO-1D trained on the same data.
- Transform tests: $\mathbf{TB} \approx \mathbf{I}_K$; orthogonality; for $k_D \equiv 1$, one HNO spectral layer with $R_n = e^{-\lambda_n^2 \Delta t}$ reproduces (19) exactly.

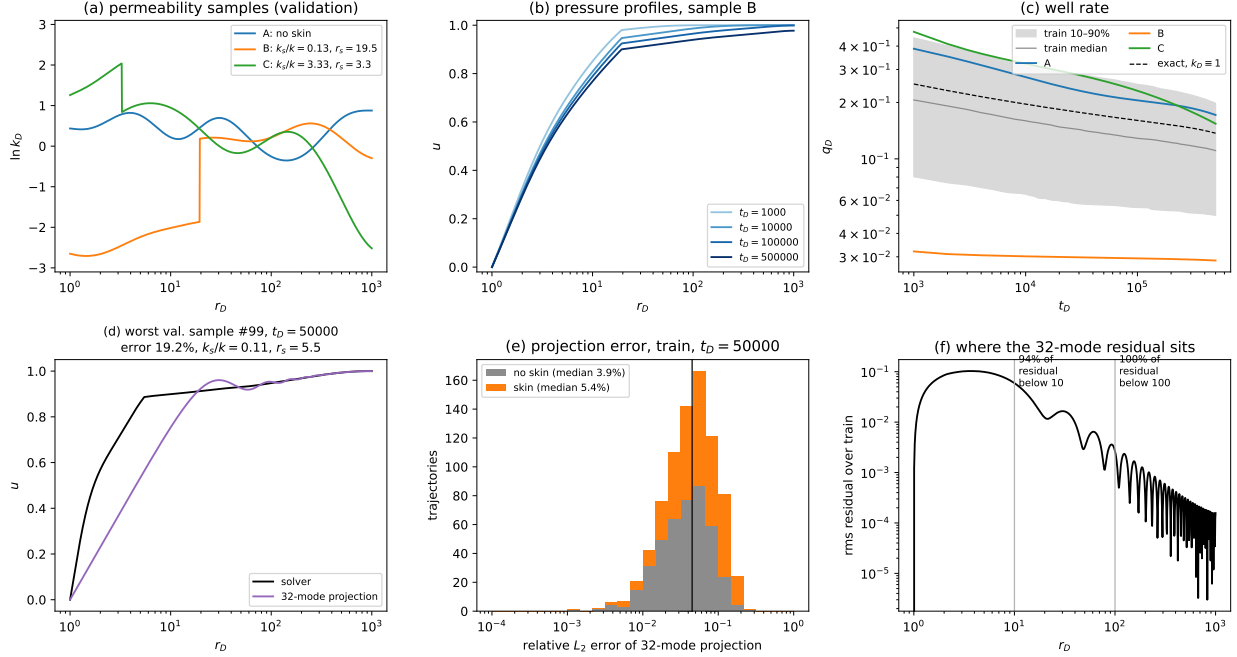


Figure 3: Dataset overview. (a) Three validation permeability fields: no skin, damaged well, stimulated well. (b) Pressure profiles of the damaged-well sample. (c) Well rate of the three samples, 10–90% band and median over the training set, and the exact $k_D \equiv 1$ rate. (d) Worst validation state at $t_D = 5 \times 10^4$ and its 32-mode projection. (e) Projection error over the training set. (f) Root-mean-square projection residual versus r .

7 Summary of phase-1 choices

Item	Choice
Geometry	$r_{eD} = 1000$ (fixed)
Varying input	$k_D(r)$ only
Operator	$(u_t, \ln k_D) \mapsto u_{t+\Delta t}$
Time step	$\Delta t_D = 1000, 500$ steps to $t_D = 5 \times 10^5$ (later an input channel)
Initial state	$u = 1$ at $t_D = 0$
Data	800 / 100 / 100 trajectories, Section 5
Spectral modes	$K = 32$
Basis	annular J_0/Y_0 eigenfunctions, eq. (13)
Grid	log-spaced in r_D , $N = 1024$
Transform	least squares, eq. (30)
Stack	PyTorch + SciPy

A Derivation of the homogeneous series coefficients

With $k_D \equiv 1$ and $u(r, 0) = u_0(r)$, the solution is $u(r, t) = \sum_m A_m \varphi_m(r) e^{-\lambda_m^2 t}$. Setting $t = 0$, taking the weighted inner product with φ_m , and using orthogonality gives

$$A_m = \hat{u}_m(0) = \frac{1}{N_m} \int_1^{r_{eD}} r u_0 \varphi_m dr, \quad N_m = \int_1^{r_{eD}} r \varphi_m^2 dr. \quad (32)$$

Below, $u_0 = 1$, and $\varphi = \varphi_m$, $\lambda = \lambda_m$ satisfy

$$(r\varphi')' + \lambda^2 r\varphi = 0, \quad \varphi(1) = 0, \quad \varphi'(r_{eD}) = 0. \quad (33)$$

Steps 1–3 use only (33). The Bessel form of φ_m enters only in Steps 4–5.

Step 1: numerator. By (33), $r\varphi = -\lambda^{-2}(r\varphi)'$ is an exact derivative, so

$$\int_1^{r_{eD}} r\varphi \, dr = -\frac{1}{\lambda^2} \left[r\varphi' \right]_1^{r_{eD}} = -\frac{1}{\lambda^2} \left(r_{eD} \varphi'(r_{eD}) - \varphi'(1) \right) = \frac{\varphi'(1)}{\lambda^2}, \quad (34)$$

since $\varphi'(r_{eD}) = 0$.

Step 2: normalization. Multiplying (33) by $r\varphi'$ gives

$$\frac{d}{dr} \left[\frac{1}{2} (r\varphi')^2 \right] + \lambda^2 r^2 \frac{d}{dr} \left[\frac{1}{2} \varphi^2 \right] = 0. \quad (35)$$

Integrating over $[1, r_{eD}]$, with the second term integrated by parts ($\int r^2(\varphi^2/2)' \, dr = [r^2\varphi^2/2] - \int r\varphi^2 \, dr$), yields

$$\int_1^{r_{eD}} r\varphi^2 \, dr = \frac{1}{2\lambda^2} \left[(r\varphi')^2 + \lambda^2 r^2 \varphi^2 \right]_1^{r_{eD}}. \quad (36)$$

At $r = r_{eD}$ only $\lambda^2 r_{eD}^2 \varphi(r_{eD})^2$ survives ($\varphi' = 0$). At $r = 1$ only $\varphi'(1)^2$ survives ($\varphi = 0$). Hence

$$N_m = \frac{1}{2} \left[r_{eD}^2 \varphi_m(r_{eD})^2 - \frac{\varphi'_m(1)^2}{\lambda_m^2} \right]. \quad (37)$$

Step 3: general coefficient. Combining Steps 1 and 2,

$$A_m = \frac{\varphi'_m(1)}{\lambda_m^2 N_m} = \frac{2\varphi'_m(1)}{\lambda_m^2 r_{eD}^2 \varphi_m(r_{eD})^2 - \varphi'_m(1)^2}. \quad (38)$$

This holds for any scaling of φ_m . Only the boundary values $\varphi_m(r_{eD})$ and $\varphi'_m(1)$ appear, so the interior shape drops out.

Step 4: well-face slope. For $\varphi_m(r) = J_0(\lambda_m r)Y_0(\lambda_m) - Y_0(\lambda_m r)J_0(\lambda_m)$, using $J'_0 = -J_1$, $Y'_0 = -Y_1$ and the Wronskian $J_1(x)Y_0(x) - J_0(x)Y_1(x) = 2/(\pi x)$,

$$\varphi'_m(1) = -\lambda_m \left[J_1(\lambda_m)Y_0(\lambda_m) - Y_1(\lambda_m)J_0(\lambda_m) \right] = -\frac{2}{\pi}, \quad (39)$$

which is the same for every mode. Substituting,

$$N_m = \frac{r_{eD}^2}{2} \varphi_m(r_{eD})^2 - \frac{2}{\pi^2 \lambda_m^2}, \quad A_m = -\frac{2}{\pi \lambda_m^2 N_m} = -\frac{4\pi}{\pi^2 \lambda_m^2 r_{eD}^2 \varphi_m(r_{eD})^2 - 4}, \quad (40)$$

in agreement with Section 3.1.

Step 5: closed form. Let $x_m = \lambda_m r_{eD}$. The eigenvalue condition (13) gives $Y_0(\lambda_m) = J_0(\lambda_m)Y_1(x_m)/J_1(x_m)$, and the Wronskian at x_m then gives

$$\varphi_m(r_{eD}) = \frac{J_0(\lambda_m)}{J_1(x_m)} \left[J_0(x_m)Y_1(x_m) - Y_0(x_m)J_1(x_m) \right] = -\frac{2 J_0(\lambda_m)}{\pi x_m J_1(x_m)}. \quad (41)$$

Substituting into Step 4,

$$\boxed{N_m = \frac{2}{\pi^2 \lambda_m^2} \left(\frac{J_0(\lambda_m)^2}{J_1(\lambda_m r_{eD})^2} - 1 \right), \quad A_m = \frac{\pi J_1(\lambda_m r_{eD})^2}{J_1(\lambda_m r_{eD})^2 - J_0(\lambda_m)^2}} \quad (42)$$

Since $N_m > 0$, $|J_0(\lambda_m)| > |J_1(\lambda_m r_{eD})|$, so $A_m < 0$, consistent with the sign of φ_m near the well. For $r_{eD} = 1000$, $A_1 = -0.25310$, $A_2 = -0.036538$, $A_3 = -0.024609$, which agree with direct quadrature to seven significant digits.